

Transactions of the American Society of Civil Engineers, vol. LXVIII, Sept. 1910, Start/End Papers

Various

An abstract geometric pattern composed of various shapes including triangles, squares, rectangles, and circles, some of which are filled with colors like red, blue, and green, while others are outlined in black or white. The pattern is dense and covers the entire background of the page.

Project Gutenberg

**Transactions of the American Society of
Civil Engineers, vol. LXVIII, Sept. 1910,
Start/End Papers**

Various



Project Gutenberg

The Project Gutenberg eBook of Transactions of the American Society of Civil Engineers, vol. LXVIII, Sept. 1910, Start/End Papers

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Transactions of the American Society of Civil Engineers, vol. LXVIII, Sept. 1910, Start/End Papers

Author: Various

Release date: June 28, 2013 [eBook #43055]

Most recently updated: October 23, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/43055

Credits: Produced by Juliet Sutherland, Martin Mayer and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK
TRANSACTIONS OF THE AMERICAN SOCIETY OF CIVIL
ENGINEERS, VOL. LXVIII, SEPT. 1910, START/END PAPERS ***

TRANSACTIONS

OF THE

AMERICAN SOCIETY

OF

CIVIL ENGINEERS

(INSTITUTED 1852)

VOL. LXVIII

SEPTEMBER, 1910

Edited by the Secretary, under the direction of the Committee on Publications. Reprints from this publication, which is copyrighted, may be made on condition that the full title of Paper, name of Author, and page reference are given.

NEW YORK

PUBLISHED BY THE SOCIETY

1910

Entered according to Act of Congress, in the year 1910, by the AMERICAN SOCIETY OF CIVIL ENGINEERS, in the Office of the Librarian of Congress, at Washington.

NOTE.—This Society is not responsible, as a body, for the facts and opinions advanced in any of its publications.

CONTENTS

THE NEW YORK TUNNEL EXTENSION OF THE PENNSYLVANIA RAILROAD

Transcriber's Note: Click on the following links to *Project Gutenberg*[™] to open each paper in a new window.

NO.		PAGE
<u>1150</u>	<u>THE NEW YORK TUNNEL EXTENSION OF THE PENNSYLVANIA RAILROAD. By Charles W. Raymond</u>	1
<u>1151</u>	<u>THE NORTH RIVER DIVISION. By Charles M. Jacobs</u>	32
<u>1152</u>	<u>THE EAST RIVER DIVISION. By Alfred Noble</u>	62
<u>1153</u>	<u>MEADOWS DIVISION AND HARRISON TRANSFER YARD. By E. B. Temple</u>	75
<u>1154</u>	<u>THE BERGEN HILL TUNNELS. By F. Lavis</u>	84
<u>1155</u>	<u>THE NORTH RIVER TUNNELS. By B. H. M. Hewett and W. L. Brown</u>	152
<u>1156</u>	<u>THE TERMINAL STATION-WEST. By B. F. Cresson, Jr.</u>	303
<u>1157</u>	<u>THE SITE OF THE TERMINAL STATION. By George C. Clarke</u>	340

1158	THE CROSS-TOWN TUNNELS. By James H. Brace and Francis Mason	391
1159	THE EAST RIVER TUNNELS. By James H. Brace, Francis Mason, and S. H. Woodard	419

MEMOIRS OF DECEASED MEMBERS

	PAGE
JOHN FISKE BARNARD, M. AM. SOC. C. E.	479
ROBERT L. ENGLE, M. AM. SOC. C. E.	480
CHARLES HERBERT DEANS, ASSOC. M. AM. SOC. C. E.	482
WILLIAM MEIER, ASSOC. M. AM. SOC. C. E.	485

This Volume and the succeeding Volume of Transactions (Vol. LXIX) will contain all the Papers descriptive of the New York Extension of the Pennsylvania Railroad. A general index covering both Volumes will be issued in Vol. LXIX.

Transcriber's Note: Click on the plate number or name to open a large view of the plate in a new window. Click on the paper number to open the paper in a new window. Click on the page number to open the paper in a new window and jump to the page indicated. (All links are to *Project Gutenberg™*.)

PLATES

PLATE		PAPER	PAGE
<u>I.</u>	<u>Map and Profile, Pennsylvania Tunnel & Terminal R. R., North Bergen Tunnel to Long Island City</u>	<u>1150</u>	<u>19</u>
<u>II.</u>	<u>Map and Profile, Harrison Yard to Bergen Hill Tunnel</u>	<u>1150</u>	<u>21</u>
<u>III.</u>	<u>Plan of Sunnyside Yard</u>	<u>1150</u>	<u>23</u>
<u>IV.</u>	<u>View of Tunnel Under Part of Atlantic Avenue, Brooklyn, N.Y.</u>	<u>1151</u>	<u>37</u>
<u>V.</u>	<u>New York Underground Railway Company: Section Through Surface and Underground Stations</u>	<u>1151</u>	<u>39</u>
<u>VI.</u>	<u>Pennsylvania Railroad Extension: Map Showing Proposed Lines Leading to Those Finally Adopted</u>	<u>1151</u>	<u>45</u>
<u>VII.</u>	<u>Plan, Profile, and Triangulation, North River Tunnels</u>	<u>1151</u>	<u>47</u>
<u>VIII.</u>	<u>Typical Sections Between Manholes, Bergen Hill Tunnels</u>	<u>1151</u>	<u>51</u>
<u>IX.</u>	<u>Map of Manhattan Island from Twenty-third to Fortieth Streets</u>	<u>1152</u>	<u>63</u>
<u>X.</u>	<u>Manhattan Shaft, Lines A and B</u>	<u>1152</u>	<u>65</u>
<u>XI.</u>	<u>Long Island Shaft, Lines A and B</u>	<u>1152</u>	<u>67</u>
<u>XII.</u>	<u>Typical Tunnel Sections</u>	<u>1152</u>	<u>69</u>
<u>XIII.</u>	<u>Plan and Profile, East River Tunnels</u>	<u>1152</u>	<u>71</u>

<u>XIV.</u>	<u>Map and Profile, Cross-Town Tunnels</u>	<u>1152</u>	<u>73</u>
<u>XV.</u>	<u>Plan and Profile of Lines A and B, and Sunnyside Yard</u>	<u>1152</u>	<u>74*</u>
<u>XVI.</u>	<u>Plan and Profile of the Pennsylvania Tunnel & Terminal R. R., from Harrison, N. J., to the Hudson River</u>	<u>1153</u>	<u>77</u>
<u>XVII.</u>	<u>Plan of Harrison Yard</u>	<u>1153</u>	<u>79</u>
<u>XVIII.</u>	<u>Details of Shelters and Platforms, Harrison Transfer Station</u>	<u>1153</u>	<u>81</u>
<u>XIX.</u>	<u>Details of Shelters and Platforms, Harrison Transfer Station</u>	<u>1153</u>	<u>81</u>
<u>XX.a</u>	<u>Fig. 1. Lift Rail and Locking Device for Hackensack River Draw-Bridge</u>	<u>1153</u>	<u>83</u>
<u>XX.b</u>	<u>Fig. 2. Lift Rail and Locking Device for Hackensack River Draw-Bridge</u>	<u>1153</u>	<u>83</u>
<u>XXI.a</u>	<u>Fig. 1. Hackensack Portal, Bergen Hill Tunnels; Method of Using Cross-Section Rod; and Belt Conveyor for Handling and Placing Concrete</u>	<u>1154</u>	<u>85</u>
<u>XXI.b</u>	<u>Fig. 2. Hackensack Portal, Bergen Hill Tunnels; Method of Using Cross-Section Rod; and Belt Conveyor for Handling and Placing Concrete</u>	<u>1154</u>	<u>85</u>
<u>XXI.c</u>	<u>Fig. 3. Hackensack Portal, Bergen Hill Tunnels; Method of Using Cross-Section Rod; and Belt Conveyor for Handling and Placing Concrete</u>	<u>1154</u>	<u>85</u>

<u>XXI.d</u>	<u>Fig. 4. Hackensack Portal, Bergen Hill Tunnels; Method of Using Cross-Section Rod; and Belt Conveyor for Handling and Placing Concrete</u>	<u>1154</u>	<u>85</u>
<u>XXII.a</u>	<u>Fig. 1. Scaffold Car; Headhouse; and Round Holes in Concrete Forms</u>	<u>1154</u>	<u>87</u>
<u>XXII.b</u>	<u>Fig. 2. Scaffold Car; Headhouse; and Round Holes in Concrete Forms</u>	<u>1154</u>	<u>87</u>
<u>XXII.c</u>	<u>Fig. 3. Scaffold Car; Headhouse; and Round Holes in Concrete Forms</u>	<u>1154</u>	<u>87</u>
<u>XXII.d</u>	<u>Fig. 4. Scaffold Car; Headhouse; and Round Holes in Concrete Forms</u>	<u>1154</u>	<u>87</u>
<u>XXIII.</u>	<u>Record of Drilling, Air Pressure, Mucking, etc., in Bergen Hill Tunnels</u>	<u>1154</u>	<u>95</u>
<u>XXIV.a</u>	<u>Fig. 1. Belt Conveyor for Handling and Placing Concrete; Water-Proofing, Portion of Completed Sand-Wall, etc.; and Methods of Placing Concrete in Forms and Bench-Walls</u>	<u>1154</u>	<u>119</u>
<u>XXIV.b</u>	<u>Fig. 2. Belt Conveyor for Handling and Placing Concrete; Water-Proofing, Portion of Completed Sand-Wall, etc.; and Methods of Placing Concrete in Forms and Bench-Walls</u>	<u>1154</u>	<u>119</u>
<u>XXIV.c</u>	<u>Fig. 3. Belt Conveyor for Handling and Placing Concrete; Water-Proofing, Portion of Completed Sand-Wall, etc.; and Methods of Placing Concrete in Forms and Bench-Walls</u>	<u>1154</u>	<u>119</u>

<u>XXIV.d</u>	<u>Fig. 4. Belt Conveyor for Handling and Placing Concrete; Water-Proofing, Portion of Completed Sand-Wall, etc.; and Methods of Placing Concrete in Forms and Bench-Walls</u>	<u>1154</u>	<u>119</u>
<u>XXV.a</u>	<u>Fig. 1. Telephone and Telegraph Ducts and Mandrels; Tunnel Lining Forms; Placing Water-Proofing; and Section of Completed Lining</u>	<u>1154</u>	<u>129</u>
<u>XXV.b</u>	<u>Fig. 2. Telephone and Telegraph Ducts and Mandrels; Tunnel Lining Forms; Placing Water-Proofing; and Section of Completed Lining</u>	<u>1154</u>	<u>129</u>
<u>XXV.c</u>	<u>Fig. 3. Telephone and Telegraph Ducts and Mandrels; Tunnel Lining Forms; Placing Water-Proofing; and Section of Completed Lining</u>	<u>1154</u>	<u>129</u>
<u>XXV.d</u>	<u>Fig. 4. Telephone and Telegraph Ducts and Mandrels; Tunnel Lining Forms; Placing Water-Proofing; and Section of Completed Lining</u>	<u>1154</u>	<u>129</u>
<u>XXVI.a</u>	<u>Fig. 1. Form of Circuit-Breaker Chamber and Traveling Gantry; Forms for Storage Chamber; Rock Packing Over Arches; Method of Water-Proofing in Timbered Tunnels, etc.</u>	<u>1154</u>	<u>137</u>
<u>XXVI.b</u>	<u>Fig. 2. Form of Circuit-Breaker Chamber and Traveling Gantry; Forms for Storage Chamber; Rock Packing</u>	<u>1154</u>	<u>137</u>

	<u>Over Arches; Method of Water-Proofing in Timbered Tunnels, etc.</u>		
<u>XXVI.c</u>	<u>Fig. 3. Form of Circuit-Breaker Chamber and Traveling Gantry; Forms for Storage Chamber; Rock Packing Over Arches; Method of Water-Proofing in Timbered Tunnels, etc.</u>	<u>1154</u>	<u>137</u>
<u>XXVI.d</u>	<u>Fig. 4. Form of Circuit-Breaker Chamber and Traveling Gantry; Forms for Storage Chamber; Rock Packing Over Arches; Method of Water-Proofing in Timbered Tunnels, etc.</u>	<u>1154</u>	<u>137</u>
<u>XXVII.a</u>	<u>Fig. 1. Method of Placing Water-Proofing and Keying Arch; View of Completed Tunnel; General View of Completed Hackensack Tunnel and Arches Through Cut-and-Cover Section; and View of Hackensack Approach</u>	<u>1154</u>	<u>145</u>
<u>XXVII.b</u>	<u>Fig. 2. Method of Placing Water-Proofing and Keying Arch; View of Completed Tunnel; General View of Completed Hackensack Tunnel and Arches Through Cut-and-Cover Section; and View of Hackensack Approach</u>	<u>1154</u>	<u>145</u>
<u>XXVII.c</u>	<u>Fig. 3. Method of Placing Water-Proofing and Keying Arch; View of Completed Tunnel; General View of Completed Hackensack Tunnel and Arches Through Cut-and-Cover</u>	<u>1154</u>	<u>145</u>

	<u>Section; and View of Hackensack Approach</u>		
<u>XXVII.d</u>	<u>Fig. 4. Method of Placing Water-Proofing and Keying Arch; View of Completed Tunnel; General View of Completed Hackensack Tunnel and Arches Through Cut-and-Cover Section; and View of Hackensack Approach</u>	<u>1154</u>	<u>145</u>
<u>XXVIII.</u>	<u>Plan and Profile of Parts of North River Tunnels</u>	<u>1155</u>	<u>153</u>
<u>XXIX.a</u>	<u>Fig. 1. Weehawken Shaft; and Tunnel Shield Showing Hood</u>	<u>1155</u>	<u>159</u>
<u>XXIX.b</u>	<u>Fig. 2. Weehawken Shaft; and Tunnel Shield Showing Hood</u>	<u>1155</u>	<u>159</u>
<u>XXX.</u>	<u>Yard and Offices at Manhattan and Weehawken Shafts</u>	<u>1155</u>	<u>161</u>
<u>XXXI.</u>	<u>Tunneling Shield</u>	<u>1155</u>	<u>167</u>
<u>XXXII.</u>	<u>Thirty-Second Street Tunnels and Shield Chambers</u>	<u>1155</u>	<u>175</u>
<u>XXXIII.</u>	<u>General Methods of Excavation Adopted for Land Tunnels</u>	<u>1155</u>	<u>177</u>
<u>XXXIV.</u>	<u>Plan and Longitudinal Section of Weehawken Tunnels</u>	<u>1155</u>	<u>193</u>
<u>XXXV.a</u>	<u>Fig. 1. Apparatus for Cleaning and Rodding Electric Cable Ducts</u>	<u>1155</u>	<u>205</u>
<u>XXXV.b</u>	<u>Fig. 2. Apparatus for Cleaning and Rodding Electric Cable Ducts</u>	<u>1155</u>	<u>205</u>

<u>XXXVI.</u>	<u>Diagram Showing Lines and Grades in River Tunnel North</u>	<u>1155</u>	<u>229</u>
<u>XXXVII.</u>	<u>Cross-Section of Subaqueous Tunnels Showing Reinforcement</u>	<u>1155</u>	<u>231</u>
<u>XXXVIII.a</u>	<u>Fig. 1. Back of Shield in South Tunnel; and Rear View of Shield During Erection of First Ring of Iron Lining</u>	<u>1155</u>	<u>237</u>
<u>XXXVIII.b</u>	<u>Fig. 2. Back of Shield in South Tunnel; and Rear View of Shield During Erection of First Ring of Iron Lining</u>	<u>1155</u>	<u>237</u>

* Transcriber's note: 75 in original.

PLATE		PAPER	PAGE
<u>XXXIX.a</u>	<u>Fig. 1. View of Meeting of Shields; and Lowering Segment into Tunnel Invert</u>	<u>1155</u>	<u>253</u>
<u>XXXIX.b</u>	<u>Fig. 2. View of Meeting of Shields; and Lowering Segment into Tunnel Invert</u>	<u>1155</u>	<u>253</u>
<u>XL.</u>	<u>Sections of Tunnel During Construction, Showing Shield, Air Locks, Platforms, Etc.</u>	<u>1155</u>	<u>255</u>
<u>XLI.a</u>	<u>Fig. 1. Placing Key Segment; and Method of Grouting Outside Iron</u>	<u>1155</u>	<u>257</u>
<u>XLI.b</u>	<u>Fig. 2. Placing Key Segment; and Method of Grouting Outside Iron</u>	<u>1155</u>	<u>257</u>
<u>XLII.</u>	<u>Duct Bench Concrete Form in River Tunnels</u>	<u>1155</u>	<u>283</u>
<u>XLIII.a</u>	<u>Fig. 1. End of Portable and Adjustable Forms for Building Side Bench; and</u>	<u>1155</u>	<u>285</u>

	<u>Steel Rod Reinforcement in River Tunnels</u>		
<u>XLIII.b</u>	<u>Fig. 2. End of Portable and Adjustable Forms for Building Side Bench; and Steel Rod Reinforcement in River Tunnels</u>	<u>1155</u>	<u>285</u>
<u>XLIV.</u>	<u>Details of 20-Ft. Movable Form and Traveler for Concrete Arches</u>	<u>1155</u>	<u>287</u>
<u>XLV.</u>	<u>Traveling Concrete Form for Face of Bench Walls</u>	<u>1155</u>	<u>291</u>
<u>XLVI.</u>	<u>Mechanical Analysis of Sand Used in Mortar and Concrete</u>	<u>1155</u>	<u>297</u>
<u>XLVII.a</u>	<u>Fig. 1. Views Showing Condition of Work at Site of Terminal Station</u>	<u>1156</u>	<u>307</u>
<u>XLVII.b</u>	<u>Fig. 2. Views Showing Condition of Work at Site of Terminal Station</u>	<u>1156</u>	<u>307</u>
<u>XLVII.c</u>	<u>Fig. 3. Views Showing Condition of Work at Site of Terminal Station</u>	<u>1156</u>	<u>307</u>
<u>XLVII.d</u>	<u>Fig. 4. Views Showing Condition of Work at Site of Terminal Station</u>	<u>1156</u>	<u>307</u>
<u>XLVIII.a</u>	<u>Fig. 1. Views Showing Excavation and Supports for Ninth Avenue Structures</u>	<u>1156</u>	<u>317</u>
<u>XLVIII.b</u>	<u>Fig. 2. Views Showing Excavation and Supports for Ninth Avenue Structures</u>	<u>1156</u>	<u>317</u>
<u>XLVIII.c</u>	<u>Fig. 3. Views Showing Excavation and Supports for Ninth Avenue Structures</u>	<u>1156</u>	<u>317</u>
<u>XLVIII.d</u>	<u>Fig. 4. Views Showing Excavation and Supports for Ninth Avenue Structures</u>	<u>1156</u>	<u>317</u>

<u>XLIX.a</u>	<u>Fig. 1. Views Showing Underpinning for Ninth Avenue Elevated Railway.</u>	<u>1156</u>	<u>321</u>
<u>XLIX.b</u>	<u>Fig. 2. Views Showing Underpinning for Ninth Avenue Elevated Railway.</u>	<u>1156</u>	<u>321</u>
<u>XLIX.c</u>	<u>Fig. 3. Views Showing Underpinning for Ninth Avenue Elevated Railway.</u>	<u>1156</u>	<u>321</u>
<u>XLIX.d</u>	<u>Fig. 4. Views Showing Underpinning for Ninth Avenue Elevated Railway.</u>	<u>1156</u>	<u>321</u>
<u>L.a</u>	<u>Fig. 1. Views Showing Condition of Work Between Ninth and Tenth Avenues, and Progress on Concrete Walls</u>	<u>1156</u>	<u>325</u>
<u>L.b</u>	<u>Fig. 2. Views Showing Condition of Work Between Ninth and Tenth Avenues, and Progress on Concrete Walls</u>	<u>1156</u>	<u>325</u>
<u>L.c</u>	<u>Fig. 3. Views Showing Condition of Work Between Ninth and Tenth Avenues, and Progress on Concrete Walls</u>	<u>1156</u>	<u>325</u>
<u>L.d</u>	<u>Fig. 4. Views Showing Condition of Work Between Ninth and Tenth Avenues, and Progress on Concrete Walls</u>	<u>1156</u>	<u>325</u>
<u>LI.a</u>	<u>Fig. 1. Views Showing Box Drains and Tie-Rods in Walls; the Completed Tenth Avenue Portal; and the Disposal Trestle</u>	<u>1156</u>	<u>333</u>
<u>LI.b</u>	<u>Fig. 2. Views Showing Box Drains and Tie-Rods in Walls; the Completed Tenth Avenue Portal; and the Disposal Trestle</u>	<u>1156</u>	<u>333</u>
<u>LI.c</u>	<u>Fig. 3. Views Showing Box Drains and Tie-Rods in Walls; the Completed Tenth Avenue Portal; and the Disposal Trestle</u>	<u>1156</u>	<u>333</u>

<u>LII.d</u>	<u>Fig. 4. Views Showing Box Drains and Tie-Rods in Walls; the Completed Tenth Avenue Portal; and the Disposal Trestle</u>	<u>1156</u>	<u>333</u>
<u>LII.a</u>	<u>Fig. 1. Girders Under the Ninth Avenue Elevated Railroad; Method of Supporting Elevated Railway Columns; and View of Inside of Form for Walls, Showing Drains, Tie-Rods, Etc.</u>	<u>1156</u>	<u>335</u>
<u>LII.b</u>	<u>Fig. 2. Girders Under the Ninth Avenue Elevated Railroad; Method of Supporting Elevated Railway Columns; and View of Inside of Form for Walls, Showing Drains, Tie-Rods, Etc.</u>	<u>1156</u>	<u>335</u>
<u>LII.c</u>	<u>Fig. 3. Girders Under the Ninth Avenue Elevated Railroad; Method of Supporting Elevated Railway Columns; and View of Inside of Form for Walls, Showing Drains, Tie-Rods, Etc.</u>	<u>1156</u>	<u>335</u>
<u>LII.d</u>	<u>Fig. 4. Girders Under the Ninth Avenue Elevated Railroad; Method of Supporting Elevated Railway Columns; and View of Inside of Form for Walls, Showing Drains, Tie-Rods, Etc.</u>	<u>1156</u>	<u>335</u>
<u>LIII.</u>	<u>Pennsylvania Station, New York City; Plan Showing Area at Track Level</u>	<u>1157</u>	<u>341</u>
<u>LIV.</u>	<u>Diagram Showing Widths of Base of Retaining Wall Required for Different Batters and Pressures, Pennsylvania Station</u>	<u>1157</u>	<u>349</u>

<u>LV.a</u>	<u>Fig. 1. Material Trestle Over N. Y. C. & H. R. R. R. Co.'s Tracks; and Construction of Pier No. 72, North River</u>	<u>1157</u>	<u>361</u>
<u>LV.b</u>	<u>Fig. 2. Material Trestle Over N. Y. C. & H. R. R. R. Co.'s Tracks; and Construction of Pier No. 72, North River</u>	<u>1157</u>	<u>361</u>
<u>LV.c</u>	<u>Fig. 3. Material Trestle Over N. Y. C. & H. R. R. R. Co.'s Tracks; and Construction of Pier No. 72, North River</u>	<u>1157</u>	<u>361</u>
<u>LVI.a</u>	<u>Fig. 1. Material Trestle Showing First Chutes in Operation; and Views of East and West Pits at Terminal Site</u>	<u>1157</u>	<u>363</u>
<u>LVI.b</u>	<u>Fig. 2. Material Trestle Showing First Chutes in Operation; and Views of East and West Pits at Terminal Site</u>	<u>1157</u>	<u>363</u>
<u>LVI.c</u>	<u>Fig. 3. Material Trestle Showing First Chutes in Operation; and Views of East and West Pits at Terminal Site</u>	<u>1157</u>	<u>363</u>
<u>±LVII.</u>	Methods of Excavation, Cross-Town Tunnels, Manhattan	<u>1158</u>	<u>393</u>
<u>LVIII.a</u>	<u>Fig. 1. Views of Parts of First Avenue Plant, Cross-Town Tunnels</u>	<u>1158</u>	<u>395</u>
<u>LVIII.b</u>	<u>Fig. 2. Views of Parts of First Avenue Plant, Cross-Town Tunnels</u>	<u>1158</u>	<u>395</u>
<u>LVIII.c</u>	<u>Fig. 3. Views of Parts of First Avenue Plant, Cross-Town Tunnels</u>	<u>1158</u>	<u>395</u>
<u>LVIII.d</u>	<u>Fig. 4. Views of Parts of First Avenue Plant, Cross-Town Tunnels</u>	<u>1158</u>	<u>395</u>

<u>LIX.a</u>	<u>Fig. 1. Methods of Tunneling, Timbering, and Lining, Cross-Town Tunnels</u>	<u>1158</u>	<u>399</u>
<u>LIX.b</u>	<u>Fig. 2. Methods of Tunneling, Timbering, and Lining, Cross-Town Tunnels</u>	<u>1158</u>	<u>399</u>
<u>LIX.c</u>	<u>Fig. 3. Methods of Tunneling, Timbering, and Lining, Cross-Town Tunnels</u>	<u>1158</u>	<u>399</u>
<u>LIX.d</u>	<u>Fig. 4. Methods of Tunneling, Timbering, and Lining, Cross-Town Tunnels</u>	<u>1158</u>	<u>399</u>
<u>LX.a</u>	<u>Fig. 1. Methods of Timbering and Underpinning, Cross-Town Tunnels</u>	<u>1158</u>	<u>407</u>
<u>LX.b</u>	<u>Fig. 2. Methods of Timbering and Underpinning, Cross-Town Tunnels</u>	<u>1158</u>	<u>407</u>
<u>LX.c</u>	<u>Fig. 3. Methods of Timbering and Underpinning, Cross-Town Tunnels</u>	<u>1158</u>	<u>407</u>
<u>LX.d</u>	<u>Fig. 4. Methods of Timbering and Underpinning, Cross-Town Tunnels</u>	<u>1158</u>	<u>407</u>
<u>LXI.</u>	<u>Sections Showing Method of Excavating and Timbering in Heavy Ground, Three-Track Tunnel; and Carriage Form for Side Walls, Cross-Town Twin Tunnels</u>	<u>1158</u>	<u>413</u>
<u>LXII.a</u>	<u>Fig. 1. Methods of Water-Proofing and Concreting, Cross-Town Tunnels</u>	<u>1158</u>	<u>417</u>
<u>LXII.b</u>	<u>Fig. 2. Methods of Water-Proofing and Concreting, Cross-Town Tunnels</u>	<u>1158</u>	<u>417</u>

<u>LXII.c</u>	<u>Fig. 3. Methods of Water-Proofing and Concreting, Cross-Town Tunnels</u>	<u>1158</u>	<u>417</u>
<u>LXII.d</u>	<u>Fig. 4. Methods of Water-Proofing and Concreting, Cross-Town Tunnels</u>	<u>1158</u>	<u>417</u>
<u>LXIII.</u>	<u>Methods of Excavation in All Rock, East River Tunnels</u>	<u>1159</u>	<u>423</u>
<u>LXIV.a</u>	<u>Fig. 1. Tunneling in Compressed Air, Air-Lock, Caisson, etc., East River Tunnels</u>	<u>1159</u>	<u>425</u>
<u>LXIV.b</u>	<u>Fig. 2. Tunneling in Compressed Air, Air-Lock, Caisson, etc., East River Tunnels</u>	<u>1159</u>	<u>425</u>
<u>LXIV.c</u>	<u>Fig. 3. Tunneling in Compressed Air, Air-Lock, Caisson, etc., East River Tunnels</u>	<u>1159</u>	<u>425</u>
<u>LXIV.d</u>	<u>Fig. 4. Tunneling in Compressed Air, Air-Lock, Caisson, etc., East River Tunnels</u>	<u>1159</u>	<u>425</u>
<u>LXV.a</u>	<u>Fig. 1. Shields Fitted with Sectional Sliding Hoods and Sliding Extensions and with Fixed Hoods and Fixed Extensions to Floors, East River Tunnels</u>	<u>1159</u>	<u>433</u>
<u>LXV.b</u>	<u>Fig. 2. Shields Fitted with Sectional Sliding Hoods and Sliding Extensions and with Fixed Hoods and Fixed Extensions to Floors, East River Tunnels</u>	<u>1159</u>	<u>433</u>
<u>LXVI.a</u>	<u>Fig. 1. Rear of Shield Showing Complete Fittings, and Shield with Lower Portion</u>	<u>1159</u>	<u>435</u>

	<u>of Bulkhead Removed, East River Tunnels</u>		
<u>LXVI.b</u>	<u>Fig. 2. Rear of Shield Showing Complete Fittings, and Shield with Lower Portion of Bulkhead Removed, East River Tunnels</u>	<u>1159</u>	<u>435</u>
<u>LXVII.</u>	<u>Methods of Tunneling in Rock, East River Tunnels</u>	<u>1159</u>	<u>437</u>
<u>LXVIII.</u>	<u>Operation of Shields, East River Tunnels</u>	<u>1159</u>	<u>439</u>
<u>LXIX.</u>	<u>Operations in Shoving the Shield Forward in Rock and Sand, East River Tunnels</u>	<u>1159</u>	<u>441</u>
<u>LXX.a</u>	<u>Fig. 1. Small Shaft, Breasting and Poling, Shutters on Front of Shield, and Hydraulic Erector, East River Tunnels</u>	<u>1159</u>	<u>443</u>
<u>LXX.b</u>	<u>Fig. 2. Small Shaft, Breasting and Poling, Shutters on Front of Shield, and Hydraulic Erector, East River Tunnels</u>	<u>1159</u>	<u>443</u>
<u>LXX.c</u>	<u>Fig. 3. Small Shaft, Breasting and Poling, Shutters on Front of Shield, and Hydraulic Erector, East River Tunnels</u>	<u>1159</u>	<u>443</u>
<u>LXX.d</u>	<u>Fig. 4. Small Shaft, Breasting and Poling, Shutters on Front of Shield, and Hydraulic Erector, East River Tunnels</u>	<u>1159</u>	<u>443</u>
<u>LXXI.</u>	<u>Operations in Shoving the Shield Forward in Sand, and Final Breasting and Bulkheading, East River Tunnels</u>	<u>1159</u>	<u>445</u>
<u>LXXII.</u>	<u>Method of Operating Shields in Soft Ground, East River Tunnels</u>	<u>1159</u>	<u>453</u>

<u>LXXIII.a</u>	<u>Fig. 1. Reinforcement of Broken Plates, and Inflow of Soft Clay Through Shield, East River Tunnels</u>	<u>1159</u>	<u>455</u>
<u>LXXIII.b</u>	<u>Fig. 2. Reinforcement of Broken Plates, and Inflow of Soft Clay Through Shield, East River Tunnels</u>	<u>1159</u>	<u>455</u>
<u>LXXIII.c</u>	<u>Fig. 3. Reinforcement of Broken Plates, and Inflow of Soft Clay Through Shield, East River Tunnels</u>	<u>1159</u>	<u>455</u>
<u>LXXIII.d</u>	<u>Fig. 4. Reinforcement of Broken Plates, and Inflow of Soft Clay Through Shield, East River Tunnels</u>	<u>1159</u>	<u>455</u>
<u>LXXIV.</u>	<u>Methods of Placing Concrete Inside the Iron Tube, East River Tunnels</u>	<u>1159</u>	<u>475</u>

[†](#) **Transcriber's note:** This plate is not in the *Project Gutenberg™* archive.

MEMOIRS OF DECEASED MEMBERS.

JOHN FISKE BARNARD, M. Am. Soc. C. E. [1](#)

[1](#) Memoir prepared by W. K. Barnard, Assoc. M. Am. Soc. C. E.

DIED FEBRUARY 6TH, 1910.

John Fiske Barnard was born in Worcester, Mass., on April 23d. 1829. He was graduated from the Bridgewater Normal School, and from Rensselaer Polytechnic Institute.

In November, 1850, Mr. Barnard entered the railway service. He held various engineering and operating positions with the Grand Trunk Railway and its subsidiary lines in Lower Canada, and served as Chief Engineer of the Grand Trunk South of the St. Lawrence River for the last three years of his connection with that road.

In May, 1869, he went to the Missouri Valley Railroad as Superintendent and Chief Engineer. During the same year he was appointed Chief Engineer of the Kansas City-St. Joseph and Council Bluffs Railroad, and remained with this road and the Hannibal and St. Joseph Railroad (both now a part of the Burlington Missouri Lines), as Chief Engineer, General Superintendent, and General Manager, until 1886. During this time Mr. Barnard was also President of the Atchison Union Depot Company and the St. Joseph Union

Stock Yards Company, Secretary and Treasurer of the St. Joseph Depot Company, and Director in various railroad companies.

In 1886, Mr. Barnard was appointed President and General Manager of the Ohio and Mississippi Railway, which position he occupied until 1892. From that time until 1893, he was engaged on several reports of projected railroads and appraisals of industrial and railroad properties.

From 1893 to 1898, Mr. Barnard was Receiver of the Omaha and St. Louis (now Wabash) Railway, during part of which time he was also President of the Alton Bridge Company, and Receiver of the St. Clair-Madison and St. Louis Belt Line.

In the spring of 1905 he moved to Los Angeles, Cal., where he lived until February 6th, 1910, when, after an illness of several months, he died at his home at the age of 81 years.

Mr. Barnard was elected a Member of the American Society of Civil Engineers on September 1st, 1880. He was also a Member of the American Geographical Society.

ROBERT L. ENGLE, M. Am. Soc. C. E. ²

² Memoir prepared by O. E. Selby, Jun. Am. Soc. C. E.

DIED OCTOBER 16TH, 1909.

Robert L. Engle was born on December 5th, 1846. He was a product of the time when opportunities for technical training were few, so that his

engineering education was gained largely by contact with actual work. He began his professional career after the Civil War, in which he served for two years in the One Hundred and Forty-eighth Indiana Volunteer Infantry.

The first construction work of any note on which Mr. Engle was engaged was the building of the Ohio and Mississippi Railroad, now a part of the Baltimore and Ohio System, extending from Cincinnati to St. Louis. Later, and up to 1878, he was connected with the construction of the Cincinnati Southern Railway, Cincinnati's municipally owned railway, as Division Engineer, at Ray Springs, Tenn. This work included several tunnels and other heavy work in the mountain territory.

After the completion of the Cincinnati Southern, Mr. Engle went West, and was engaged in the construction of the Santa Fé Railroad at Trinidad, Colo., in the capacity of Assistant Chief Engineer. Under his direction the Royal Gorge Hanging Bridge was built, and much other interesting work was carried out. Mr. Engle's forte was location, and in the mountainous regions of the West he found ample exercise for this faculty. From Trinidad he moved to Santa Fé, N. Mex. His name is borne by Engle, N. Mex., now a thriving Western town.

While still in the West, Mr. Engle was connected with the Mexican Central Railroad, at Chihuahua, Mexico, on construction work, and with the Denver and Rio Grande Railroad. It is thus seen that he played a part in much of the important pioneer railroad development of the mountain region of the West.

In 1885, Mr. Engle began work on the location and construction of the Chicago, Burlington, and Northern Railroad, now part of the Burlington System, being located at St. Paul, Minn., as Assistant Chief Engineer of that

portion of the line north of La Crosse, Wis. Later, his jurisdiction included the whole line. During 1887 and a part of 1888, he had charge of the construction of the Illinois Valley and Northern Railroad, as Chief Engineer, at La Salle, Ill. In the latter part of 1888 he conducted surveys for coal branch lines connecting with the Chesapeake and Ohio Railway, in the mountains of West Virginia.

Beginning in September, 1889, Mr. Engle was Resident Engineer on the construction of the Louisville and Jeffersonville Bridge over the Ohio River. During his term of service the substructure, involving several deep pneumatic foundations, was built, and parts of the approaches were erected. During his stay at Louisville Mr. Engle was selected as Arbitrator in a matter of disputed classification between the company and the contractor for the Pike's Peak Rack Railroad, and effected a satisfactory settlement. Among other things his later service included location work on the Tennessee Central Railroad, in 1892; location and construction work for the Missouri, Kansas and Texas Railroad, in Arkansas; and construction work on the Tidewater Railroad, now the Virginian Railway, at Princeton, W. Va. At the time of his death he was employed as Engineer for the contracting firm of Carpenter and Boxley, at Johnson City, Tenn.

With the death of Robert L. Engle, the Profession loses one of those sturdy, self-made engineers, to whom the country is largely indebted for pushing railroad construction overland and through the West. In character, rugged like the mountains with which he was associated, he was still the gentlest of souls to those associated with him in subordinate capacities. The writer knew him as Chief and friend for many years, and cannot recall any

departures from the lines of the highest dignity, rectitude, good habits, and good nature.

Mr. Engle was a Member of the Engineers' Club of Cincinnati from the time of its organization. He maintained his home in Cincinnati for twenty-one years, while his engagements kept him at various other places.

On February 20th, 1879, Mr. Engle married Miss Sallie McQueety, of Cincinnati, and is survived by her and their son and two daughters. His family and social relations were most happy, although his enforced absences from home kept him from much of the social contact which his qualities deserved.

Mr. Engle was elected a Member of the American Society of Civil Engineers on September 7th, 1881.

CHARLES HERBERT DEANS, Assoc. M. Am. Soc. C. E. [3](#)

[3](#) Memoir prepared by Emil Diebitsch and Edwin S. Jarrett, Members,
Am. Soc. C. E.

DIED MARCH 7TH, 1909.

Charles Herbert Deans was born in Chester, Delaware County, Pa., on November 30th, 1863, and died at his home in Phoenixville, Pa., on March 7th, 1909.

His father, Charles Woodbury Deans, was prominent in educational work, and was active in the early organization and in the popularization of the

Common School System of the State of Pennsylvania.

On his father's side Mr. Deans was descended from the Deans and Sterling families, who, immediately following the War of the Revolution, emigrated from Connecticut to Susquehanna and Wyoming Counties, Pennsylvania. His mother was Priscilla Lyons Williams, of Chester, Delaware County, Pa., who was descended from the Lyons family of New Jersey and the Williams and Pennell families of Pennsylvania.

From both his father and his mother Mr. Deans inherited a taste and aptitude for study. His youthful environment was among books and in an atmosphere which naturally encouraged the desire he early formed to fit himself for a professional life.

His education was begun in private schools, but later he attended the public schools, and was graduated from the High School at Phoenixville, Pa., in 1881. He spent the next four years in practical work, learning business methods, becoming an excellent and accurate accountant, and familiarizing himself, in the works of the Phoenix Iron Company, with mill and shop methods and practice, and the metallurgy of iron and steel.

In 1885, Mr. Deans entered Lehigh University, well prepared in his studies, with a mind ripe for the absorption of further knowledge, and a temperament for enjoying to the utmost the four years of University life before him. He was a good student, standing well up in the first quarter of his class. He was elected a member of Theta Delta Chi Fraternity, was one of the Editors and Assistant Business Manager of the college Annual in his Junior year, and Business Manager of the Engineering Journal in his Senior year. He was graduated in 1889 with the degree of C. E.

As a boy Mr. Deans was fond of games and all healthy outdoor sports. He was a lover of Nature and of animals, fond of fishing and hunting, and was never happier than when roaming the beautiful woods and mountains of his native State. With such tastes it was natural to find him, in his college days, a participator in, and an enthusiastic supporter of, athletic games.

Not only in athletics, but in all things pertaining to Lehigh University, Mr. Deans was a most loyal and enthusiastic son of his Alma Mater, both at college and after he had gone out into the world. He thoroughly appreciated the benefits derived from his technical training, and was so eager that others should share them, that early in his business career he advanced sufficient funds to two ambitious young men to carry them through Lehigh.

Immediately after graduation Mr. Deans entered the employ of SooySmith and Company, the well-known foundation engineers and contractors. He rapidly advanced to positions of responsibility with this company, and, in 1895, became its Vice-President and Chief Executive Officer. When, a year or two later, Charles SooySmith, M. Am. Soc. C. E., retired from active business, Mr. Deans organized, from the SooySmith and Company staff, the Engineering Contract Company, of which he became President. Pressure of business seriously undermining his health, he was forced to give up temporarily all work in 1900, and to spend the next two years in the mountains of Northern Pennsylvania. On regaining his health, he associated himself with the firm of John Monks and Son, of New York City, and, at the time of his death, he was Second Vice-President of that company.

While under his executive charge, both SooySmith and Company and the Engineering Contract Company, constructed a number of the most

important bridge foundations in the United States, and the former firm first successfully introduced pneumatic work in the foundations of the modern high office buildings of New York City, notably the Manhattan Life, Washington Life, Standard Oil, and Empire Buildings on Lower Broadway. At the time of his death, Mr. Deans was in full charge of the building of the piers of the reconstructed Baltimore and Ohio bridge over the Susquehanna River, at Havre de Grace, Md.

In the early years of his connection with SooySmith and Company, Mr. Deans was employed on work in the field, rising from subordinate positions to that of Superintendent in responsible charge of work. During this period he acquired an intimate and practical knowledge of foundation construction, and his subsequent career gave evidence of the value of this training. Being thus well-equipped, Mr. Deans soon became notable as a business engineer. His judgment on all substructure engineering problems was quick and keen, his thorough technical knowledge being supplemented by his penetrating practical sense. His business ability was of a high order, and his efficiency was largely increased by his industry and methodical habits. As a negotiator, he was in the first rank. His quick appreciation of the essentials in business transactions, his fertile resource in the most complicated financial dealings, his patience and persistence in the face of discouragement or delay, and his inflexible determination when once his decisions were reached, were qualities which placed him in the highest rank as a contracting engineer. To those with whom he came in close contact, Mr. Deans will always be remembered as exemplifying the ideal combination of technical training with business efficiency.

He lightened the seriousness of his business transactions with a quick sense of fun, a fondness for a good story, and an infectious good humor. His genuine interest in the work of his associates and his unfeigned delight in their success won him many friendships which lasted throughout his life and which now keep his memory warm in the hearts of those who were fortunate enough to know him intimately. Strong of will, keen and clear-sighted in business transactions, loyal to his friends and to the interests entrusted to him, he was, above all, a genial, honorable, many-sided man, who loved his fellow men.

Mr. Deans leaves a mother, Mrs. Charles W. Deans, of Phoenixville, Pa., a brother, John Sterling Deans, M. Am. Soc. C. E., Chief Engineer of the Phoenix Bridge Company, and two sisters, Mrs. R. Barclay Calley, of Seattle, Wash., and Mrs. Elmer E. Keiser, of Tacony, Pa.

In 1893, Mr. Deans married Miss Helen Arnold of West Chester, Pa., who, with two sons, Charles Woodbury, aged 15, and Malcolm Arnold, aged 13, survives him.

Mr. Deans was elected a Junior of the American Society of Civil Engineers, on December 3d, 1890, and an Associate Member on May 6th, 1896.

WILLIAM MEIER, Assoc. M. Am. Soc. C. E. [4](#)

[4](#) Memoir prepared by William A. Theodorsen and E. James Fucik,
Associate Members, Am. Soc. C. E.

DIED FEBRUARY 14TH, 1910.

William Meier, the son of the Reverend Jacob L. and Mary Meier, was born in Muscatine, Iowa, on April 10th, 1878, the family moving to Chicago, Ill., in the same year.

Mr. Meier received his education in the public schools of Chicago and at the University of Illinois, from which he was graduated in 1901, with the degree of B. S. in Civil Engineering.

After his graduation, Mr. Meier was engaged with various firms, principally in bridge and structural work. For a time he was with William M. Hughes, M. Am. Soc. C. E., and in January, 1905, he entered the service of the Scherzer Rolling Lift Bridge Company, as Assistant in the Chicago office; and later was appointed Assistant Engineer and Eastern Representative, with headquarters in New York City. At the time of his death, Mr. Meier was employed in the Bridge Department of the Chicago and North Western Railway.

On February 14th, 1910, in diving from a spring-board, in the natatorium of the Young Men's Christian Association, he struck his head against the side or bottom of the tank. When his body was taken from the water, life was extinct, and all efforts at resuscitation were futile.

Mr. Meier took great interest in all that pertained to his profession.

He was elected an Associate Member of the American Society of Civil Engineers, on June 1st, 1909. He was also a Member of the Western Society of Engineers.

*** END OF THE PROJECT GUTENBERG EBOOK TRANSACTIONS
OF THE AMERICAN SOCIETY OF CIVIL ENGINEERS, VOL. LXVIII,
SEPT. 1910, START/END PAPERS ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a

lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work.

The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs

1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format

must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT

LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including

checks, online payments and credit card donations. To donate, please visit:
www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.